

On the Impossibility of Reproducing Quantum Correlations with a Real-Valued Mother Function $Z(x)$: A Constraint on Future Extensions

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Abstract

This short note reports a negative but informative result concerning the so-called *Mother Function* $Z(x) = x^5 \ln(x) \sin(2\pi/x) e^{-x/x_p}$, proposed as part of the Sundus Unified Theory ($Z + C + A = 1$). We construct a local hidden-variable model directly from $Z(x)$ and test it against the CHSH inequality. The model yields $S = 1.82$, respecting Bell's local bound $S \leq 2$. We prove that this is not a numerical accident: any model in which the measurement outcome depends only on a shared real variable x and a real phase $\sin(2\pi/x)$ is necessarily local, and therefore cannot reproduce the quantum value $S = 2\sqrt{2}$ observed in Aspect-type experiments. This establishes a rigorous constraint: any future extension of $Z(x)$ aiming to capture quantum entanglement must possess (i) a complex-valued structure that enters the probability density and (ii) a non-factorizable bilinear form $Z(x_1, x_2) \neq Z(x_1)Z(x_2)$. The numerical value $S_Z = 1.82$ is not a discovery; it is the expected consequence of Bell's theorem applied to a local hidden-variable model.

Keywords: Bell's theorem, CHSH inequality, quantum entanglement, local hidden variables, Mother Function, Sundus Theory.

1 Introduction

The *Sundus Unified Theory* proposes $Z + C + A = 1$, in which $Z(x)$ plays the role of structural generator. We ask whether $Z(x)$ can encode quantum entanglement without non-local signalling. This note answers in the negative for the real-valued form and shows it is a necessary consequence of Bell's theorem.

2 Preliminaries

$$Z(x) = x^5 \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/x_p}, \quad x > 0 \quad (1)$$

x is dimensionless (in units of l_p). $x_p = 1.22$ was used for this numerical evaluation only, as a dimensionless scale parameter; no physical unit is claimed.

$$\phi(x) := 2\pi/x, \quad m_a(x) = \text{sign}[\sin(\phi(x) + a)], \quad p(x) = \frac{|Z(x)|^2}{\int |Z|^2 dx} \quad (2)$$

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3 Construction

$$E_Z(a, b) = \int_0^\infty p(x) m_a(x) m_b(x) dx \quad (3)$$

This is a local hidden-variable model: outcome depends only on shared x and local setting.

θ (deg)	$E_{\text{QM}} = -\cos \theta$	E_Z (this work)	$ diff $
0	-1.0000	+1.0000	2.0000
22.5	-0.9239	+1.0000	1.9239
45	-0.7071	+0.9989	1.7060
67.5	-0.3827	+0.9882	1.3709
90	0.0000	+0.9104	0.9104
112.5	+0.3827	+0.5055	0.1228
135	+0.7071	-0.5674	1.2745
157.5	+0.9239	-0.9997	1.9236
180	+1.0000	-1.0000	2.0000

Table 1: Non-circular test: $x \in [0.05, 30]$, 2×10^5 points, $S_Z = 1.82$

4 Main Result

$S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$, $a = 0, a' = \pi/2, b = \pi/4, b' = -\pi/4$. $S_Z = 1.82 < 2$. Any model $E(a, b) = \int p(x) m_a(x) m_b(x) dx$ satisfies $|S| \leq 2$ (Bell-CHSH). The numerical value $S_Z = 1.82$ is not a discovery; it is the expected consequence of Bell's theorem applied to a local hidden-variable model.

5 Why it fails

(i) No antisymmetry: singlet needs $E = -\cos(a-b)$. Real $Z(x)$ symmetric. $Z_{\text{new}} = x^5 \ln(x) e^{i\phi} e^{-x/x_p}$ still gives $|Z_{\text{new}}|^2 = |Z|^2 \rightarrow \text{local}$. (ii) Factorizability: $Z(x)$ is single-variable, entanglement needs $Z(x_1, x_2) \neq Z(x_1)Z(x_2)$.

6 Necessary conditions

(i) Complex structure entering $p(x)$, not only amplitude. (ii) Non-factorizable: $Z(x_1, x_2) = Z(x_1)Z(x_2)e^{i\Phi(x_1, x_2)}$, $\Phi \neq \Phi_1 + \Phi_2$.

7 Conclusion

Real-valued $Z(x)$ cannot reproduce quantum correlations. $S_Z = 1.82$ is Bell's bound, not failure. This constrains future extensions.

References

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[4] Nobel Prize in Physics 2022.